CASE ANALYSIS - YATRAFY GEN-AI ASSISTANT

1. Market Analysis

1.1 Total Addressable Market

India's online travel market currently has a size of USD 23.1 billion dollars and is expected to touch 33.9 billion dollars in the next five years, indicating a 7.76% CAGR. Yatrafy currently holds 20% of this overall market share in India in the industry, and its major revenue comes from accommodation bookings. We have 944.7 million wireless data users, and this number is escalating steadily due to government backed digitization programs like National Digital Tourism Mission, etc which shows that there is a continuous shift in booking habits from offline channels to digital platforms. With increasing online travel agencies, competition amongst them is also on the rise, leading to deeper loyalty programs and dynamic pricing options. Overall, India's online travel market is growing exponentially.

1.2 Serviceable Available Market

This section pertains to the segment of Indian travellers travelling for work or leisure purposes, who will be our target consumers. According to a new report by the Boston Consulting Group (BCG), India is one of the major contributors to the next wave of global leisure travel growth, thanks to its rising middle class and younger population. Post COVID-19, India quickly bounced back on its travel wagon, which is indicated by its moderate to strong growth in leisure travel spending from 2019 to 2024, and it is expected to continue to grow. Indian travelers are increasingly exploring both their own country and the world. Domestic, regional, and international trips are expected to rise by 3%, 4%, and 6%, respectively, per year. BCG's survey found that Millennials, Gen-Zers and Gen-Xers plan more trips than the older population. This section of the population is tech-savvy and socially conscious. Another prominent section of travel is pilgrimages. Combination of work trips with leisure also makes a major chunk of the travel market. More than 70% of travellers in countries such as China, India, Nigeria, and Saudi Arabia plan to combine work trips with leisure.

1.2 Serviceable Obtainable Market

Metric	Value	Source
TAM (Online Travel Market)	USD 23.1 B	Cited from market analysis report [1]
Leisure and Work Travel Share of Online Market	70%	Research indicates leisure travel accounts for approximately 70% of online travel spending in India[1]
Yatrafy's Market Share	20%	Yatrafy's reported overall market share in India (as of 2025).
Target GenAI Adoption Rate	60%	Based on consumer surveys showing that 60% online travel bookers are likely to use AI for search and itinerary generation [4]

The SOM is calculated based on a two-step approach: calculating the size of the SAM then applying Yatrafy's existing market share and the target adoption rate of the GenAI tool.

SOM Calculation:

1. SAM (Online Leisure Travel Market):
   SAM = TAM x Leisure and Work Travel Share
   SAM = $23.1 Billion x 0.70 = $16.17 Billion
2. SOM (Winnable Share):
   SOM = SAM x Yatrafy's Market Share x GenAI Adoption Rate
   SOM = $16.17 Billion x 0.20 x 0.60 = $1.94Billion

From the above calculations we can say that SOM is estimated to be $1.94 Billion. This is an achievable target because:

- Yatrafy is focusing only on the 20% of online market it already owns, removing the high cost and difficulty of customer acquisition.
- Indian travelers are open to using GenAI, with 60% of online bookers expressing a likelihood to use AI for planning

2. Customer segmentation

In the Market Analysis section, we analysed the market and understood whom to target. In this section, we will be classifying our target consumers based on two factors. This segmentation will help us focus on specific user behaviors so we can develop product features that resonate with different user needs and pain points.

2.1 Segment A - Why do they want to travel?

This segmentation by purpose of travel helps distinguish different decision journeys and success criteria. This enables the assistant to tailor itineraries, constraints, and communication style to each use case.

- Work trip - this comprises of work related trips for a variety of purposes
  Examples:
  - Executives meeting with clients
  - Business owners travelling for research purposes
  - Employees being sent onsite to work on a project
- Leisure trip - this comprises of trips taken purely for pleasure
  Examples:
  - Solo/ Group trips to explore a new place
  - Pilgrimages for religious fulfillment
  - Trips taken for a new learning experience, like going to Thailand to learn Muay Thai

2.1 Segment B - How do they want to travel?

This segmentation by travel style reflects differences in budget, risk tolerance and expectations from the trip. This helps the GenAI assistant to personalize recommendations, filters, and upsell opportunities while staying aligned with each traveler's preferences.

1. Budget travellers - they travel with very limited items and stay in affordable places. Their aim is to travel and explore with limited resources.
2. Luxury travellers - they stay and travel in the most comfortable and luxurious ways. Their aim is to travel and explore without compromising on their needs and comforts.
3. Slow travellers - they travel with the intention of slowing down and exploring at their own pace, without any checklists or heavy itineraries.

User personas, user journey maps and user research are mentioned below as Exhibit A, B and C..

2.2 Pain Points
Pre-trip:

- Time-consuming research and planning
  - Comparing options for flights, stays, and activities across multiple sites takes a lot of time.
  - Travelers often struggle to turn vague ideas ("Europe in summer") into a workable itinerary.
- Fragmented booking experience
  - There is no unified place to book flights, hotels, local transport, activities, and add-ons together and managing bookings across different platforms becomes confusing.
- Information gaps for international travel
  - Hard to find reliable, up-to-date information on currency exchange, international SIMs/eSIMs, and local regulations.

On-trip:

- First-mile/last-mile local transport
  - Finding safe and convenient transport in the city they are in.
  - Language barriers and lack of local knowledge make it hard to compare options or avoid scams.
- Food and dietary preferences
  - Identifying restaurants that cater to specific needs (vegan, vegetarian, halal, gluten-free, Jain, etc.) is challenging, especially abroad.
- Safety and emergency support
  - Locating nearby hospitals, pharmacies, and police stations in an emergency is stressful and time-sensitive.

3. Market Gap
3.1 Competitor Overview

The online travel industry has seen rapid adoption of GenAI for trip recommendations and basic chat support, but in today's market there is no product that offers an end-to-end platform to transform ambiguous travel ideas into complete itineraries—from pre-trip planning through on-trip emergencies. In this section, we will explore the various competitors, analysis of existing products and present what gap the GenAI feature of Yatrafy will address.Key competitors in this space include global players like Google Travel, Booking.com, and Expedia, alongside Indian rivals such as MakeMyTrip's multilingual GenAI assistant, plus niche tools like Wonderplan and Tripplanner.ai. The table below analyzes the top three most relevant competitors, highlighting their strengths and critical gaps that Yatrafy's GenAI assistant is uniquely positioned to fill.

Competitor	Features	Gaps / Limitations
Google Travel / AI Mode in Search	- Integration with Search and Gmail, pulling bookings into unified timeline - AI mode plans itineraries, compares options, completes bookings in conversational flow - Ingests live prices, traffic, demand trends and predictive suggestion to generate recommendations	- Aggregates information from many different travel service providers, further actions need to be taken on the third-party platform itself - Lacks India-specific nuances (IRCTC rails, local buses)
Booking.com AI Trip Planner	- Conversational AI for planning, filtering stays/transport - AI-generated summaries of feedback, advanced filtering tools and rebooking assistance	- Accommodation-centric; limited multi-modal (flights/trains/buses) integration - Limited on-trip support (food preferences, emergencies, visas) - Friction between chat/filters/pages, not fully chat-first
MakeMyTrip GenAI Assistant	- Integrated with flights/hotels/trains/buses/packages - Simplifies discovery, personalizes by budget/duration/preferences	- Discovery-focused, does not provide end-to-end booking/post-sales support - Limited on-trip support (transfers, dietary discovery, emergencies)

3.2 Unique Selling Proposition (USP) of Yatrafy's GenAI Travel Planning Assistant

- Holistic trip management from start to finish, covering multi-modal transport, accommodations, curated local experiences, and dynamic post-booking services — including cancellations, refunds, and travel document assistance — all in one place.
- Personalized concierge features, such as recommendations for local dining according to

dietary preferences, emergency services and first/last-mile transport that reduce difficulties in unfamiliar locations.
- Provision of verified information on country-specific logistics for the itinerary's destination like currency exchange rates and local SIM/eSIM recommendations.
- Single thread booking flow that removes the need to visit multiple pages
- Local language support and direct integration with Indian travel pillars, notably IRCTC train bookings.

4. Minimum Viable Product

Yatrafy's GenAI MVP is a conversational, end-to-end travel concierge that transforms ambiguous leisure trip ideas into personalized, multi-modal itineraries, driving direct accommodation bookings via a single chat interface for Yatrafy's existing customer base.

4.1 Features

4.1.1 Functional

- Allows users to input vague, multi-constraint queries (e.g., "Adventure trip for 5 days in the Himalayas on a budget").
- The AI tailors the itinerary based on customer's preferences, which is taken through filters/conversations ensuring all suggestions (stays, transport, activities) align with the declared style.
- Allows users to search and compare real-time pricing and availability for Flights, IRCTC Trains, and major Buses relevant to the itinerary.
- Users can select and book all integrated components (accommodation, flight/train) directly from the generated itinerary within the single chat interface, eliminating the need to visit separate pages.
- Provides instant, verified information on country-specific logistics like currency exchange rates and local SIM/eSIM recommendations.
- A knowledge-base access point within the chat to instantly provide contact details and navigation to nearby hospitals, pharmacies, and police stations.
- Integrates filters for nearby restaurants that specifically cater to the user's stated dietary preferences (e.g., vegan, vegetarian, halal).

4.1.2 UI/UX

- Quick buttons at the top for trip length, number of people, food preferences (vegan/vegetarian/halal), budget, and trip mood (adventure/relax). Chat asks for more details if needed.
- As users pick flights, hotels, or activities, a side panel shows sections like "Travel," "Stays," "Plan" with summaries and edit/book buttons.
- Clean interface with photos, maps, prices, and big "Book Now" buttons

4.2 LLM Model Analysis

This section shows an analysis of several large language models (LLMs) that are capable of powering our GenAI assistant. Further analysis will help us select the optimal model based on accuracy, latency, and cost.

LLM Model	Rationale
Google Gemini 1.5 Flash	Accurately converts natural language into structured API calls for IRCTC, accommodation, and flight bookings
OpenAI GPT-4o	Ideal for creating personalized itineraries
Anthropic Claude 3.5 Sonnet	Ensures the user's travel style and personal preferences are maintained across the entire chat session
Krutrim or Sarvam-1 (OpenHathi)	Boosts local adoption by providing planning and support in Hindi and other major Indian languages
Meta LLaMA 3 / Mistral	Allows Yatrafy to train the model directly on its proprietary IRCTC data and unique accommodation inventory, leading to a highly customized and efficient product.

4.3 Success Metrics

Metric	Target Goal	Rationale
Itinerary-to-Accommodation Conversion Rate (IACR)	Higher than the platform's current conversion rate.	It tracks how many generated itineraries lead to successful bookings to verify its positive impact on revenue.
Customer Satisfaction (CSAT/NPS)	NPS > 50 for users who complete an itinerary.	Measures loyalty and organic growth. It validates that the pain points are addressed for a convenient user experience.

References

1. https://www.mordorintelligence.com/industry-reports/online-travel-market-in-india
2. https://economictimes.indiatimes.com/industry/services/travel/tourism-market-forecast-indias-rising-middle-class-joins-the-15-trillion-global-travel-opportunity-bcg-report/articleshow/122199607.cms?from=mdr
3. https://www.bcg.com/publications/2025/the-15-trillion-opportunity-in-leisure-travel
4. AI reshapes Indian travel experience with better reviews, planning; most still want full control: Reports, Fortune India/LocalCircles

AI Usage: Perplexity and Google Gemini were used for understanding how to approach this case study, conducting market research and technology analysis and for evaluating product metrics.

EXHIBIT A - User Personas

Ananya Sharma

28, Remote Program Manager

Bio

Anya is a high-achieving corporate professional who leverages her remote Program Manager role to lead a location-independent life. She is a high-frequency solo traveler who prioritizes Workations, favoring slow, immersive exploration over typical tourist checklists. She spends 1-2 weeks in each international city, balancing her synchronous work duties with a deep desire for cultural connection. As a solitary traveler, her life is governed by meticulous planning and a non-negotiable standard for personal safety and security.

Income	High (₹18-25 LPA equivalent)		
Education	Post-Graduate Degree (MBA/Equivalent)		
Status	Single, Location-Independent		
Travel Style	High-Frequency	Workation	Solo-Leisure

🎯 Goals

- Perfectly blend synchronous work productivity with slow, immersive travel.
- Achieve non-negotiable personal security via explicit, data-backed safety ratings.
- Move beyond tourist lists to unique, skill-based local experiences.
- Eliminate anxiety regarding logistical failures or support abandonment during the trip.

💥 Pain Points

- Wasting hours navigating 5+ platforms for one complex, multi-component itinerary.
- Exhausting manual process of verifying neighborhood lighting, transit, and emergency proximity.
- Struggle to find reliable, current, local information on SIMs, transfers, and currency exchanges.
- Fear of encountering unresponsive, static support during a critical on-trip crisis.

✨ Wants & Needs

- A single, conversational interface that executes all complex bookings instantly.
- Explicitly displayed, verifiable safety data integrated into all accommodation options.
- Must ensure accommodation has high-speed internet, ergonomic space, and power stability.
- A guaranteed, rapid path to human support for crisis management.

Rahul Kumar

40, Senior Corporate Manager

Bio

Ramesh is the meticulous family guardian focused on creating lasting core memories. His planning is governed by Family First (safety and happiness) and Prudence, prioritizing value over the lowest price. Given rigid financial constraints (EMIs) and fixed school holidays, he demands total efficiency, reliability, and safe execution from all bookings to ensure the trip provides a necessary, stress-free break from his corporate life.

Income	Middle Class to Upper Middle Class; high disposable income but high expenses (school fees, EMIs).		
Education	Tier - 1 University Degree		
Status	Married, 2 children		
Travel Style	Guided	Comfort-focused	Family Voyager

🎯 Goals

- Book a perfect summer vacation that satisfies the entire family.
- Maintain budget while ensuring high perceived value.
- Minimize time spent researching flights and accommodation across multiple platforms.
- Guarantee access to reliable transport and nearby emergency services (hospitals, pharmacies).

💥 Pain Points

- Frustration managing bookings for four across different, non-integrated platforms.
- Struggle to reliably find vegetarian and kid-friendly dining options in new locations.
- Overwhelmed by the volume of choices for family activities and stays.
- Fear of unsafe or dishonest "first-mile/last-mile" transport (e.g., airport taxis).

✨ Wants & Needs

- A single platform to recommend and execute all group bookings (flights, hotels, cabs).
- Automatic filtering for "Family with Kids" amenities and guaranteed "Vegetarian" food highlights.
- Immediate access to location-specific emergency information (hospitals, police) via the bot interface.
- AI-driven suggestions to shift dates or routes to maximize budget savings (e.g., "Saves ₹15,000").

EXHIBIT B - User Journey Map

STAGES	SEARCH & CROSS-PLATFORM COMPARISON	ACCOMMODATION VETTING & BOOKING	ACTIVITY RESEARCH & PURCHASE	PAYMENT & FINANCIAL STRUCTURING	LOGISTICAL CONSOLIDATION & EXTERNAL PREP	POST-TRIP ADMINISTRATION
GOALS	Find the best value flight deal (max one stop) with efficiency in mind.	Secure the only property that meets non-negotiable safety and work connectivity standards.	Find safe, enriching, solo-friendly cultural experiences that fit the schedule.	Manage costs by splitting personal (EMI) and corporate (Card/UPI) expenses seamlessly	Consolidate fragmented data, prepare for non-booking logistics (SIM, calendar).	Complete administrative tasks quickly and feed safety/ connectivity data back into the system.
ACTIONS	1. Checks price trends on Ixigo. 2. Identifies a preferred airline/ route. 3. Uses a major OTA (Goibibo) to check wallet discounts. 4. Books the optimal flight.	1. Filters hotels aggressively (Rating > 4.5, "Serviced Apartment"). 2. Manual Vetting: Opens address on Google Maps/ Street View to check neighborhood lighting and co-working proximity. 3. Reserves the vetted property.	1. Searches for cultural experiences (e.g., cooking class) on Klook. 2. Checks user reviews for "solo" or "small group" mentions. 3. Books 2-3 key experiences.	1. Reviews the total bill. 2. Decision Point: Chooses EMI (Buy Now Pay Later) for personal costs and Corporate Card/UPI for work portion.	1. Downloads 5 separate confirmation PDFs/Vouchers. 2. Manually copies flight/hotel details into her personal work calendar (adjusting for time zone difference). 3. Manual Research: Spends 45 mins searching forums for reliable eSIM/SIM card vendor.	1. Consolidates all receipts. 2. Manually separates Work vs. Leisure expenses on her corporate finance portal. 3. Leaves a brief review on the hotel for "connectivity" and "night-time safety."
THOUGHTS	"Ixigo confirmed the fare is fair. I'll book on MMT for their customer service history. That's one big thing done."	"This manual vetting takes hours, but I must do it for peace of mind. The app should know I need good connectivity near my work office."	"Klook has better focused activity options than the OTA. I trust their verified vendor badges more. The cooking class is a great way to meet people safely."	"The EMI option makes the big cost manageable upfront, but it's another loan to track. I hope the UPI transaction doesn't fail at this price point."	"I have three different apps/ emails/documents for one trip. If my flight is delayed, I have to manually update the hotel and the activity vendor."	"The trip was great, but the admin is a nightmare. There needs to be a feature that auto-categorizes this. If a crisis hits, I know the OTA will vanish."
EMOTIONS	😌 Focused	😡 Frustrated	🙂 Relieved	😐 Anxious	😟 Overwhelmed	😎 Satisfied
PAIN POINTS	• Inefficient use of time jumping between comparators and booking agents to find the real lowest price with discounts. • Hidden convenience fees surface late in the process.	• Lack of internal, granular safety data (neighborhood lighting) and reliable, tested Wi-Fi speed ratings needed for work. Necessitates mandatory platform switching.	• Experience marketplaces are not fully integrated with the flight/hotel flow. • Risk of double-booking or scheduling conflicts with the work calendar.	• High cognitive load choosing between credit, UPI, and various BNPL/EMI options, often involving external bank or fintech partner approval processes. • Fear of transaction failure due to high value.	• All essential non-booking tasks (eSIM, currency exchange, airport transfers, time zone management) require completely separate research. • No Centralized Itinerary.	• Tedious, manual receipt management and expense reporting, especially when splitting personal/corporate costs. • OTAs offer no proactive support or prompt escalation when a flight or hotel booking is cancelled or delayed.
OPPORTUNITIES	**Pitch as Productivity Tool:** Use language that appeals to her values (e.g., "Save 10+ hours of planning").	**Internal Safety Visualization:** Integrate a map layer showing public transit proximity and basic safety/lighting data. Wi-Fi Verification: Show tested Wi-Fi speeds	**Integrated Itinerary:** Automatically add activities to the main trip calendar, checking for conflicts with work hours.	**Unified Payment/Split Tool:** Offer a single gateway that automates the split between corporate and personal payments, including tracking EMI.	**"Anya's Local Logistics Brief":** Auto-generate a personalized document covering SIMs, emergency contacts, local customs, and neighborhood safety tips before departure.	**Proactive Crisis Management & Human Path:** Implement an automated system that monitors flight status, contacts hosts, and proposes solutions instantly. Dedicated Emergency Human Escalation Path (24/7 access).

STAGES	DATE & BUDGET LOCK	MULTI-PLATFORM FLIGHT SEARCH	ACCOMMODATION VETTING	ACTIVITY & DINING PLANNING	FINAL BOOKING & FINANCIAL RECONCILIATION	PRE-TRIP LOGISTICS & CRISIS MANAGEMENT
GOALS	Lock in dates and budget limits to manage yearly savings plan.	Find the optimal group flight deal (value over lowest price) with manageable layovers.	Secure family-friendly accommodation with verified amenities (pool, extra beds).	Plan reliable, safe, and vegetarian-friendly dining and engaging activities for the children.	Consolidate costs, verify passenger details, and finalize payments.	Distribute consolidated documents and ensure readiness for airport transfers and potential crises.
ACTIONS	1. Locks in fixed travel dates (school holidays). 2. Establishes a rigid, per-head daily spending limit in Excel. 3. Discusses 2-3 destination options with his wife.	1. Searches across 3-4 platforms for group flight prices. 2. Filters by layover tolerance (must be short) and checked baggage. 3. Books the deal via the OTA with the lowest fee.	1. Filters accommodation by "Family Room," "Free Breakfast," and "Pool Access." 2. Manual Vetting: Calls the hotel or checks reviews for guaranteed extra beds. 3. Books the most flexible cancellation policy.	1. Searches blogs for kid-friendly activities. 2. Cross-checks restaurants on Zomato/TripAdvisor for vegetarian and kid-friendly options. 3. Purchases 1-2 major attraction tickets.	1. Consolidates all costs into his master spreadsheet, verifying budget adherence. 2. Chooses UPI/credit card for rewards points. 3. Double-checks children's names and ages on all tickets.	1. Prints paper copies of all vouchers (precaution). 2. Sends wife a detailed WhatsApp itinerary. 3. Searches for a reliable, pre-booked airport taxi service to avoid haggling/scams.
THOUGHTS	"The dates are locked. Now the stress begins. I must adhere to the budget or the whole year's savings plan is ruined."	"The low-cost option has an 8-hour layover—that will destroy the kids. Value over lowest cost, always."	"Does 'Family Room' mean two full beds, or one bed and a sofa? I wish I didn't have to call them to confirm this."	"The kids will love the museum. Now, finding a reliable vegetarian dinner nearby is the main logistical headache."	"The spreadsheet is green! We are under budget. Let me check the cancellation window one last time."	"Airport transport is high-risk. I need a reliable sedan for safety and space. If the flight cancels, I have zero prompt support."
EMOTIONS	😟 Anxious	😠 Annoyed	😒 Frustrated	🙂 Relieved	😊 Happy	😫 Stressed
PAIN POINTS	• School holiday dates cause significant price spikes. • Time wasted creating and updating manual budget spreadsheets.	• Managing group baggage and seat selection across fragmented airline policies. • High cognitive load comparing group pricing across platforms.	• Vague Family Amenities: Lack of standardized info on room layouts, extra bed charges, and children's facility quality. • Dietary filters are unreliable or non-existent on major OTAs.	• Fragmentation of activity booking from the main itinerary. • Manual cross-referencing of dining options for specific dietary needs.	• Zero automation for expense reporting (manual data entry required). • High risk of data entry errors with four passengers' details.	• Significant fear of unsafe/dishonest "first-mile/last-mile" transport. • Too many printouts to consolidate. • OTAs provide no proactive support during flight/hotel cancellations; Ramesh must manage re-booking and compensation for the family alone.
OPPORTUNITIES	**Budget Assistant:** Offer an AI tool that sets and tracks budgets in real-time against actual bookings.	**Group Price Optimizer:** Automatically calculate and present the true cost-per-head across all platforms, including fees.	**Standardized Vetting:** Offer a verified "Family-Ready Score" that checks room layout, bed count, and specific amenities (pool safety).	**Integrated Dining Search:** Automatically suggest highly-rated vegetarian restaurants along the planned itinerary route.	**Automated Expense Report:** Generate an exportable expense file categorized by trip/passenger for corporate/personal admin.	**Proactive Crisis Management:** Implement 24/7 human escalation and an automated system that monitors flight status, proposes re-booking, and handles compensation claims instantly for the entire group.
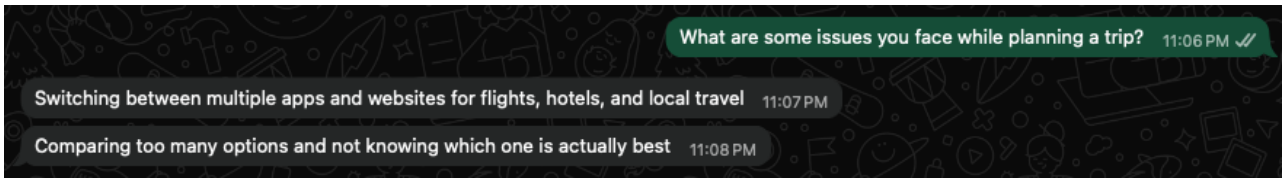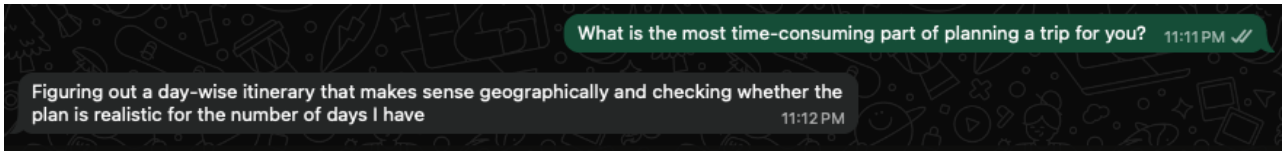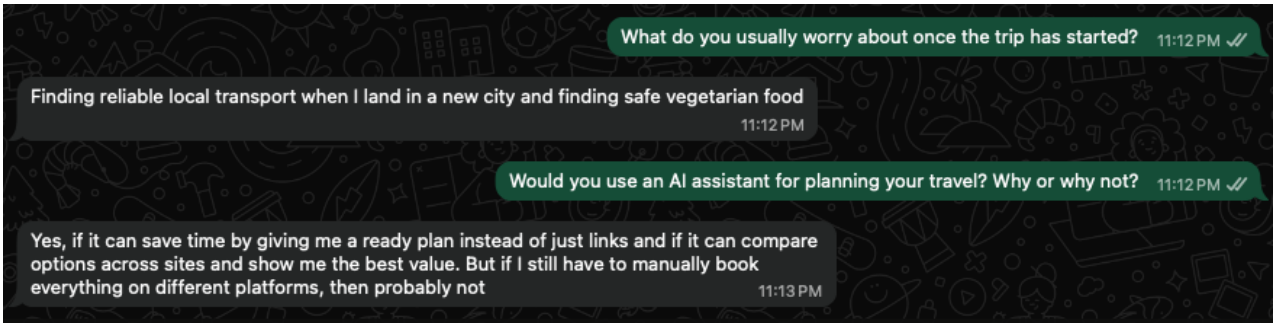
EXHIBIT C - Screenshot of user research